

Data Analytics With Tableau

1. Introduction

Project Title: Visualization Tool for Electric Vehicle Charge and Range Analysis

Team ID: SWTID-2026-7520

Team Members:

- **Team Leader:** Komal Nagendra Panchakarla
 - Managed the project by storing EV data in the MySQL database, performing SQL operations, connecting the database with Tableau, and publishing the Tableau dashboard.
 - Integrated the Tableau dashboard and story with a Flask web application, recorded the end-to-end project demonstration video, coordinated the team's activities, and completed the final project submission.
- **Team member:** Ayyappa Addala
 - Prepared the dataset for visualization by cleaning the data, handling missing values, removing duplicates, and transforming the data into a suitable format for Tableau analysis.
 - Created calculation fields in Tableau to derive meaningful metrics and developed the Tableau Story with multiple scenes to effectively present the project's insights and analysis.
- **Team member:** Abdul Saleem
 - Managed the amount of data rendered to the database by validating the imported records, ensuring data accuracy, and optimizing the data for efficient analysis and visualization.
 - Implemented and configured interactive data filters in Tableau, enabling users to dynamically analyze EV charging station data based on different regions, charger types, power ratings, and other relevant parameters.
- **Team member:** Akasapu Hema Naga Venkata Mahesh
 - Collected and organized the required Electric Vehicle datasets from reliable sources, ensuring the data was accurate, complete, and suitable for project analysis.
 - Designed and developed multiple interactive Tableau visualizations and graphs to analyze EV charging infrastructure, battery performance, and vehicle efficiency, providing meaningful insights through effective data visualization.
- **Team member:** Sumitra Somepally
 - Developed unique and interactive Tableau visualizations based on the EV datasets and designed a responsive, user-friendly dashboard that effectively presents key insights through charts, filters, and visual elements.
 - Prepared comprehensive project documentation by describing the complete step-by-step project development procedure, including data analysis, visualization creation, dashboard design, testing, and implementation.

2. Project Overview

Purpose: The purpose of the Visualization Tool for Electric Vehicle Charge and Range Analysis project is to develop an interactive business intelligence solution that analyzes electric vehicle charging infrastructure and vehicle performance using Tableau. The project aims to provide meaningful insights through data visualization, helping users understand EV charging patterns, battery performance, driving range, and charging station availability. It also integrates Tableau dashboards with a Flask-based web application to provide an interactive and user-friendly experience.

- Analyze electric vehicle charging station and vehicle performance data.
- Visualize charging patterns, battery efficiency, and driving range using interactive dashboards.
- Compare EV models based on charging time, battery capacity, range, and efficiency.
- Support data-driven decision-making through business intelligence and analytics.

Features: For Users:

- Interactive Dashboard – View EV charging station distribution, battery performance, and driving range through interactive Tableau dashboards.
- Charging Station Analysis – Analyze charging stations by region, charger type, service type, and power rating.
- EV Model Comparison – Compare electric vehicles based on battery capacity, charging time, driving range, and efficiency.
- Interactive Filters – Filter visualizations by region, charger type, service type, power rating, and other parameters for customized analysis.
- Tableau Story – Explore project insights through a step-by-step Tableau Story explaining key findings.
- Web-Based Access – Access the published Tableau dashboard and story through a Flask-based web interface.

For Administrator (Future Scope):

- Database Management – Store and manage EV datasets using MySQL for efficient data retrieval and updates.
- Real-Time Data Integration – Integrate live EV charging station and vehicle data for up-to-date analytics.
- Advanced Analytics – Implement predictive analysis to estimate battery degradation, charging demand, and driving range.
- User Management – Enable secure user authentication and role-based access for administrators and analysts.
- Dashboard Maintenance – Monitor dashboard performance, update datasets, and publish new reports for continuous analysis.

3. Architecture

(Dashboard Interface)

- Developed using Tableau dashboards for interactive EV charge and range analysis.
- Provides visualizations such as charts, graphs, KPIs, and filters.
- Allows users to analyze charging performance and vehicle range.

Data Processing Layer

- Collects EV charging datasets and performs data cleaning and preprocessing.
- Handles missing values, duplicate removal, and data transformation.
- Prepares data for Frontend analysis and visualization.

Database (MySQL)

- Stores EV charging and vehicle information.
- Main table: EV_DATA

Stored Attributes:

- Vehicle Model
- Battery Capacity
- Charging Type
- Charging Power
- Charging Time
- EV Range
- Charging Station Details

Visualization Layer (Tableau)

- Creates interactive dashboards and reports.
- Displays EV range analysis, charging efficiency, and performance insights.
- Supports filtering and exporting report

4. Setup Instructions

Prerequisites

- **MySQL – Database for storing EV charging data.**

- **Tableau Desktop** – For creating and viewing dashboards.
- **VS Code** – Recommended code editor.
- **Git** – To clone the project.

Installation Steps:

1. Clone the Project

```
git clone https://github.com/komalnagendra/Electric-Vehicle-Charge-and-Range-Analysis
cd EV-Charge-Range-Analysis
```

2. Setup Database

- Install MySQL.
- Create database EV_ANALYSIS.
- Import the EV dataset into the database.

3. Run Backend

```
cd backend
pip install -r requirements.txt
python app.py
```

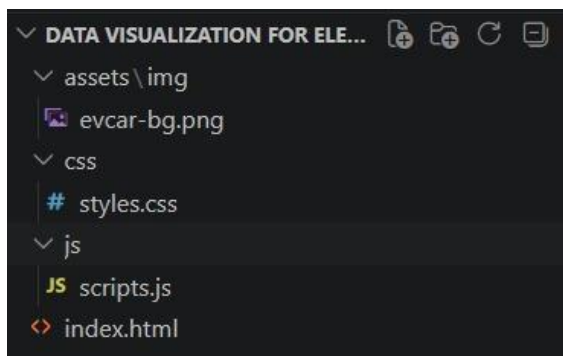
4. Open Tableau Dashboard

- Open Tableau Desktop.
- Connect to the EV database/dataset.
- Load the dashboard workbook.

Application Access:

- Dashboard: Open through Tableau Desktop / Web view.
- Backend API: <http://localhost:5000>

5. Folder Structure



Running the Application

Frontend:

The application can be executed by opening the **index.html** file in a web browser.

URL: <http://localhost:5500>

Tableau Dashboard:

- Open Tableau workbook file.
- Connect the EV dataset/database.
- View interactive EV Charge Range Analysis dashboard.

6. Authentication

How Authentication Works:

- Users access the EV Charge Range Analysis dashboard through the web application.
- The application connects to the database to retrieve EV charging and range

data.

- Only authorized users can view or analyze the dashboard (if login is enabled).

Method Used:

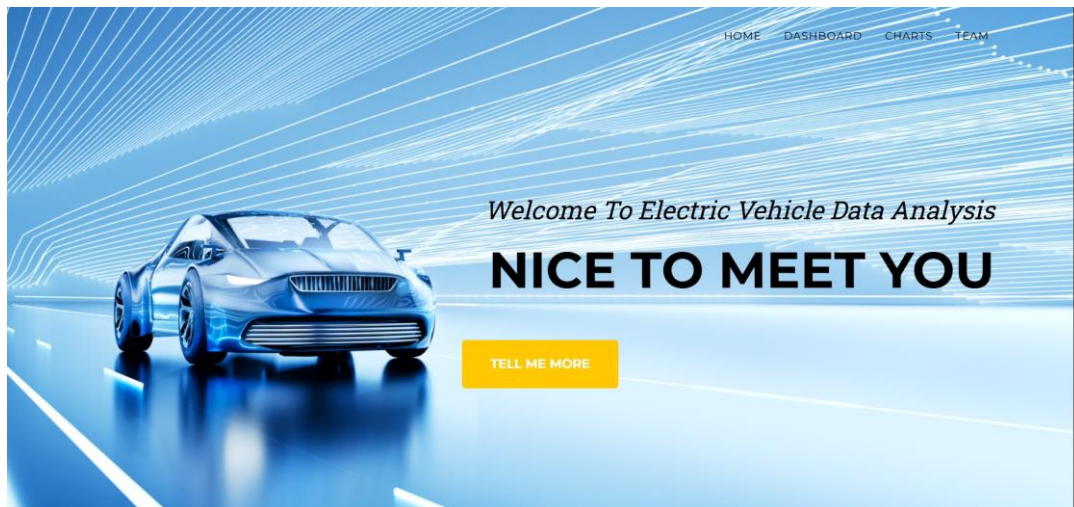
- Basic username and password authentication.
- User credentials are verified before granting access.
- The application maintains the user session during usage.

Recommendations for Improvement:

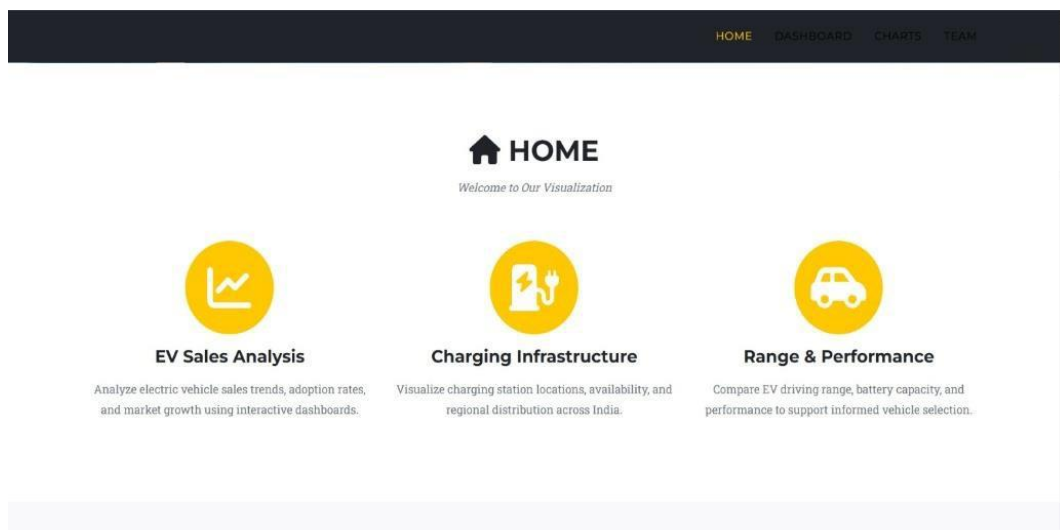
- Implement JWT (JSON Web Token) authentication.
- Encrypt passwords using bcrypt.
- Add role-based access (Admin/User).
- Use secure session management for enhanced security.

7. User Interface

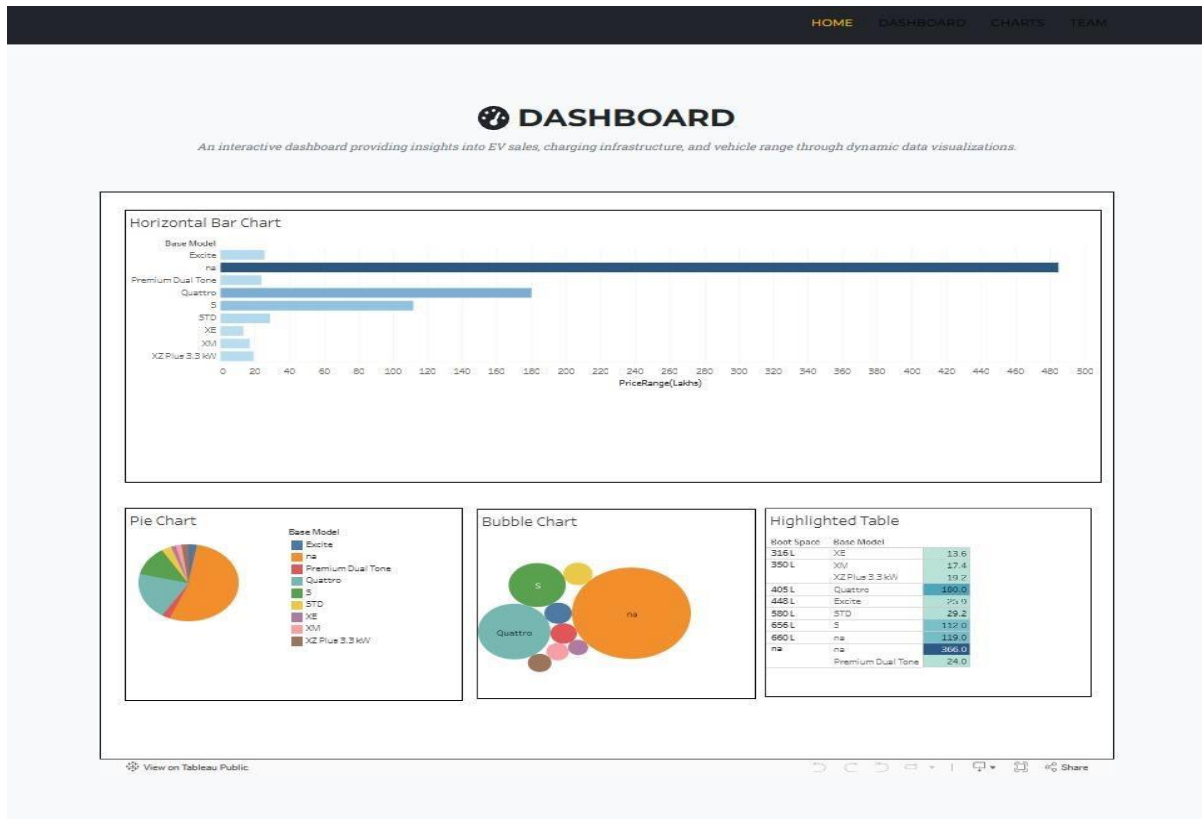
Home page:



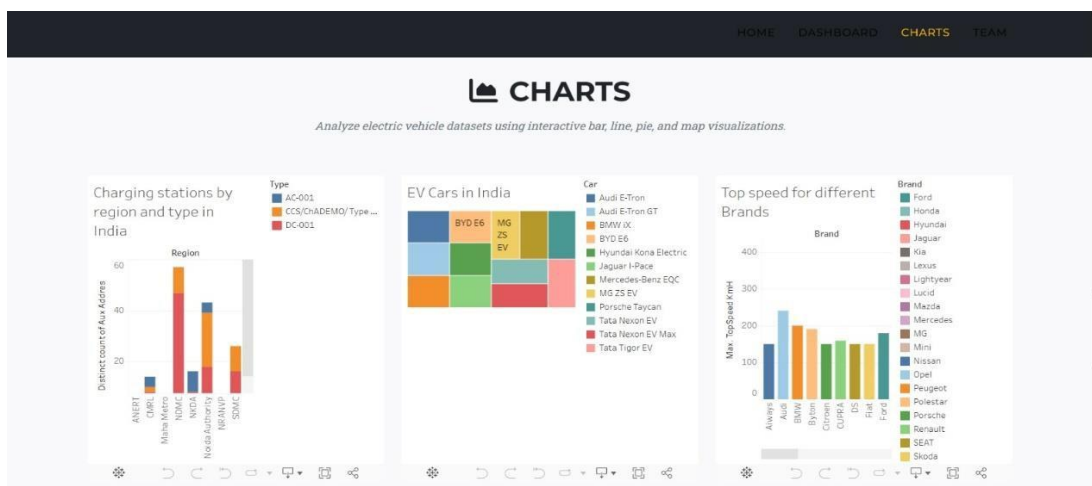
Home Features Page:



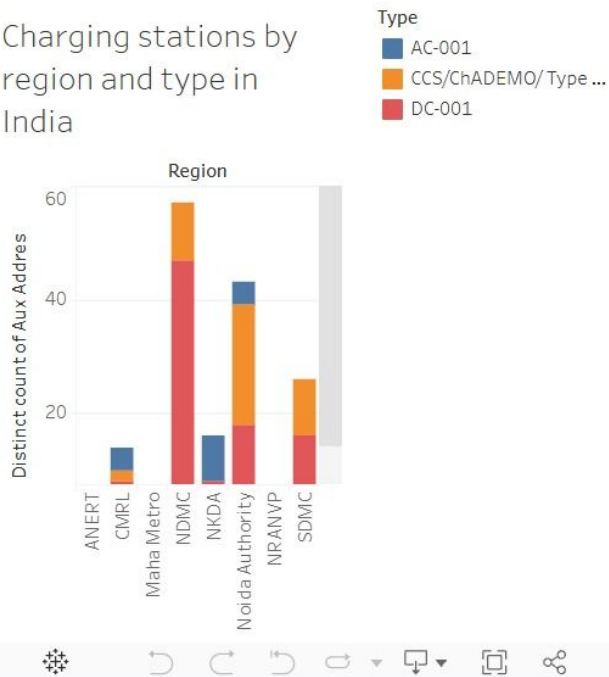
Dashboard:



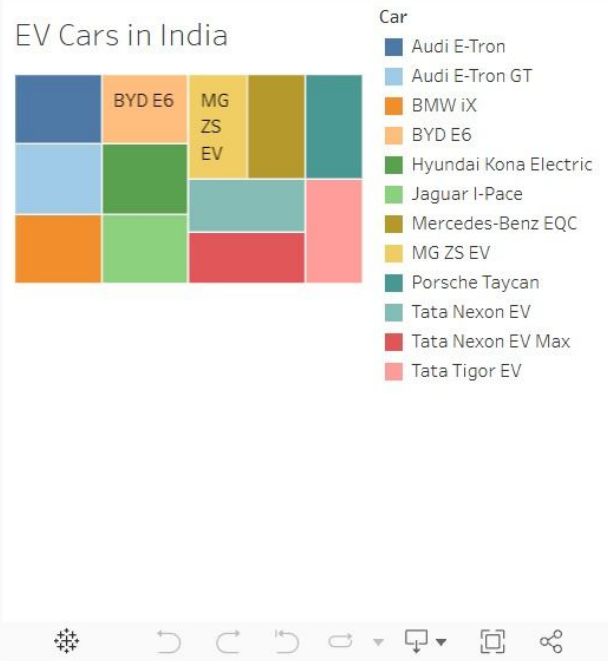
Data Visualization Page:



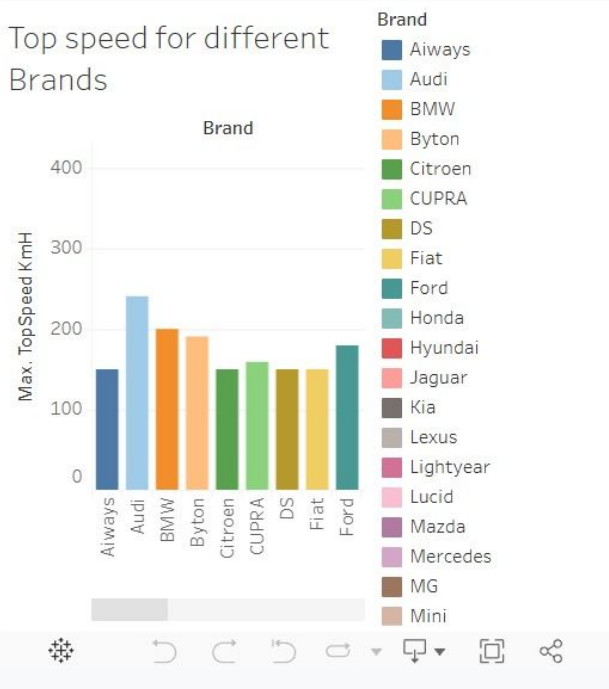
Charging stations by region and type in India



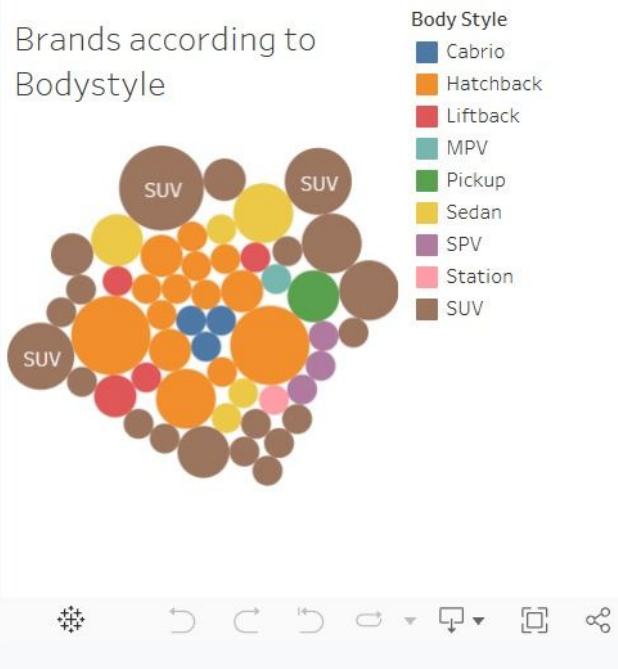
EV Cars in India



Top speed for different Brands



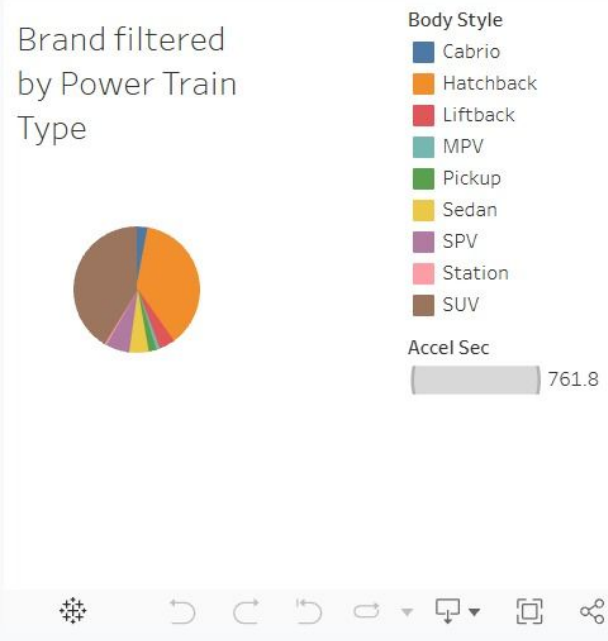
Brands according to Bodystyle



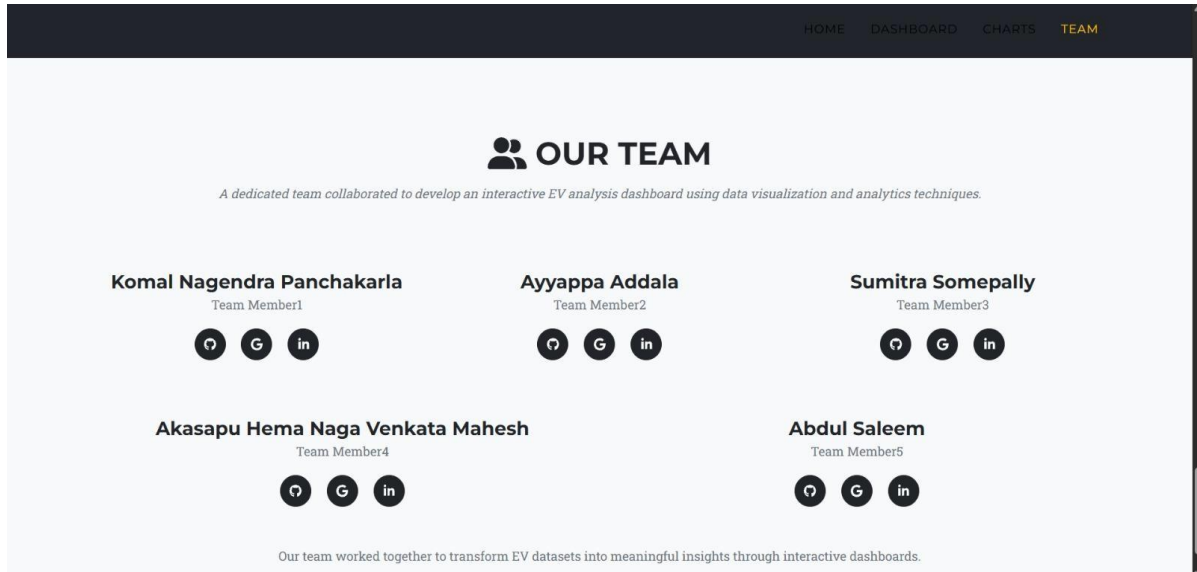
Price for different cars in India



Brand filtered by Power Train Type



Team Page:



Project Website Link:

<https://komalnagendra.github.io/Electric-Vehicle-Charge-and-Range-Analysis/>

8. Testing

- **Manual Testing:** Verified dashboard functionality, filters, charts, and report generation.
- **Database Testing:** Checked data retrieval and accuracy from the MySQL database.
- **Flask Testing:** Tested backend connectivity between the application and database.
- **Tableau Testing:** Validated dashboard interactions, visualizations, and calculated metrics.

9. Known Issues

- Dashboard depends on the accuracy of the input dataset.
- No real-time EV charging data integration.
- Analysis is limited to the available dataset.
- Testing is performed manually; automated testing is not implemented.

10. Future Enhancements

- Integrate real-time EV charging station data using APIs.
- Add predictive analytics for EV range estimation using Machine Learning.
- Develop a live monitoring dashboard.
- Enhance the dashboard with additional KPIs and interactive visualizations